FinTrust: Trustworthy Small Language Models for Financial Tasks

Heet Mehta, Krish Patel, Aryan Patel
heetmehta.work@gmail.com krishpatel19.tech@gmail.com aryanpatel26.ce@gmail.com

I. ABSTRACT

Small language models (SLMs) may be suitable for focused financial applications where the compute requirements of larger models are difficult to accommodate. Yet credible performance on financial tasks involves more than generating convincing text: datasets must be controlled, metrics must be defined consistently, errors must be examined, and experiments must be repeatable. This paper develops an evaluation pipeline for numerical reasoning and continuous sentiment regression on FinQA and FiQA Sentiment. It combines QLoRA-based parameter-efficient adaptation with dataset validation, isolated evaluation splits, prompt and sampling studies, confidence diagnostics, and inference-cost measurements. The Phase-2.1 data contract also tracks source-file fingerprints, checks FinQA reference programs with the official execution procedure, and audits exact matches and potential near-duplicate leakage.

In the available validation summary, the balanced joint-QLoRA configuration using Qwen3-1.7B (E5) records 0.6750 FinQA execution accuracy, a FiQA MAE of 0.1301, and $R^2 = 0.7301$. A separate run with Phi-4-mini-instruct (E8) reports 0.701 execution accuracy and a FiQA R^2 of 0.809. These figures refer to validation data; they are not results from the protected test sets. The E12 failure totals also disagree with the execution-accuracy value attributed to E5. The original prediction artifacts must therefore be re-evaluated to resolve this mismatch before publication. The pipeline is intended to make the experimental evidence traceable, not to treat predictive scores as proof of calibrated uncertainty or safe financial decisions.

***Index Terms*—Financial language models, small language models, FinQA, FiQA, QLoRA, data integrity, confidence estimation, trustworthy artificial intelligence.**

II. INTRODUCTION

Financial analysis tools often need to interpret prose, extract information from tables, and carry out calculations whose results can be checked. Reports include monetary amounts, ratios, percentages, comparisons across time periods, and statements that only make sense in context. A fluent response is not necessarily a correct one: a model can choose the wrong figure, apply an unsuitable operation, or produce a reasoning program that fails during execution.

Although large language models have advanced text comprehension and generation, their compute and memory requirements can limit use in constrained environments. Smaller models offer another option, particularly when targeted adaptation can improve performance without updating every parameter of the underlying model.

Using a smaller model does not remove the main reliability concerns. Output quality can change with prompt wording, arithmetic or program construction may fail partway through a solution, and a model's apparent confidence may not reflect whether its answer is correct. Experimental comparisons are also difficult to interpret when data splits overlap or results from different runs are reported as though they were directly comparable.

This work studies how to train and assess small language models for financial numerical reasoning and sentiment regression under a traceable experimental protocol. In addition to measuring task performance, the pipeline links dataset versions and training settings to saved predictions and the metrics derived from them, making the results easier to reproduce and audit.

A. Research Problem

The research question is how to adapt compact language models to financial tasks without losing experimental traceability. This requires suitable metrics for each task, clear separation between development data and protected evaluation data, and retained predictions that support error inspection and confidence analysis.

B. Research Objectives

The study addresses the following objectives:

1) Establish a reproducible dataset contract for FinQA and FiQA Sentiment.
2) Compare prompting baselines with QLoRA fine-tuning.
3) Evaluate task-specific and joint-task training configurations.
4) Investigate balanced and natural sampling strategies.
5) Analyze numerical reasoning failures and sentiment prediction errors.
6) Examine the relationship between model confidence and prediction correctness.
7) Report training resource consumption and inference latency.

C. Contributions

The paper contributes a version-controlled data preparation specification, a shared experimental workflow for two different

financial prediction tasks, comparisons between prompting and parameter-efficient fine-tuning, and analyses of model errors, confidence signals, and computational cost.

Validation and protected-test results are treated separately throughout the study. Validation data may inform model selection and configuration decisions; the protected test sets should only be used after those choices have been fixed, so that they provide a less biased estimate of final performance.

III. Related Work

A. Numerical Reasoning over Financial Documents

FinQA evaluates numerical question answering grounded in financial documents and tables, with reference solutions represented as executable programs [1]. Solving its questions requires selecting relevant evidence and applying appropriate mathematical operations, including addition, subtraction, multiplication, and division.

ConvFinQA broadens the task to multi-turn financial question answering [2]. These benchmarks illustrate that a financially correct response often depends on evidence-grounded calculations and a sequence of operations, rather than on language fluency alone.

Accordingly, this study reports execution accuracy, program accuracy, and answer accuracy separately. Each metric captures a different aspect of the model's numerical response and should not be treated as interchangeable with the others.

B. Financial Sentiment Analysis

The FiQA resource includes financial text with sentiment annotations and aspect-related information [3]. In the regression setting used here, a prediction is a numerical sentiment score, not a class label.

Keeping sentiment as a continuous target retains the annotation scale and makes it possible to quantify the size of prediction errors. The evaluation therefore uses MAE and the coefficient of determination alongside the separate metrics for numerical reasoning.

C. Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) modifies selected layers through trainable low-rank matrices while leaving the pretrained weights fixed [4]. As a result, fine-tuning updates substantially fewer parameters than full-model training.

QLoRA combines LoRA adapters with quantization of the base model weights [5]. This design is useful when GPU memory is constrained because the original weights do not need to be stored and optimized entirely in full precision.

D. Trustworthy Model Evaluation

A model's score alone is insufficient to judge whether it can be trusted in practice. Data leakage, misleading confidence signals, and deployment costs can all affect suitability. Exact-match and near-duplicate checks are useful for finding possible overlap between splits, while calibration analyses test whether confidence contains meaningful information about prediction quality.

TABLE I
Dataset sizes and split policies in the Phase-2.1 contract.

Dataset	Train	Val.	Test	Role
FinQA	6251	883	1147	Reasoning
FiQA	822	117	234	Regression
FinBen-FOMC	0	0	496	External eval.
FinTrust	–	–	Protected	Trust eval.

These checks are incorporated into the workflow. The study does not assume that fine-tuning automatically makes outputs calibrated, robust, or safe.

E. Research Gap

The research gap is a compact-model financial baseline that evaluates numerical reasoning and sentiment regression while also documenting split isolation, reproducible artifacts, error patterns, confidence behaviour, and resource use. The intended outcome is a baseline that can be independently checked before broader protected-set evaluation.

IV. Dataset and Integrity Contract

A. Dataset Selection

FinQA is the source for numerical reasoning examples, while FiQA Sentiment is used for continuous sentiment prediction with entity and aspect metadata. Because these datasets represent different tasks, their outputs are assessed with different metrics.

The pipeline reserves FinBen-FOMC for external assessment of monetary-policy stance classification. FinTrust is treated as an evaluation-only benchmark registry covering seven areas: trustfulness, robustness, safety, fairness, privacy, transparency, and knowledge discovery.

The FinQA and FiQA totals in the table are the counts expected by the Phase-2.1 contract. The final version should include the validation report produced by the actual run so readers can confirm that the processed data matched these expectations.

B. Canonical Record Design

Records from each dataset are mapped to a common schema containing identifiers, dataset and task labels, split membership, system instructions, inputs, targets, and metadata. This shared representation simplifies the training pipeline while retaining the original reference annotations.

For FinQA, the model input includes the financial context, table, and question. The target contains a structured program and executable answer:

```
PROGRAM: <program>
ANSWER: <answer>
```

Reference answers and supporting evidence remain available in the metadata, but the gold evidence is excluded from the model's input to avoid giving it information unavailable at prediction time.

For FiQA, the target retains the continuous sentiment score:

`SENTIMENT_SCORE: <number>`

The continuous score is retained rather than mapped into newly defined sentiment categories. This keeps the task aligned with the original annotations and supports direct numerical comparison with the reference values.

C. Source Provenance and Hashing

The Phase-2.1 contract records SHA-256 fingerprints for source files and exported artifacts. Canonical serialization makes structured data suitable for consistent hashing. The resulting manifests connect the source version, normalized dataset, and downstream training files.

A matching hash is not evidence that the underlying data are semantically correct. Its purpose is narrower: it helps establish that an artifact has not changed since it was recorded in the manifest, which is useful when experiments are repeated or transferred between environments.

D. Integrity Gates

Before training, the pipeline applies pass/fail checks to dataset counts, required columns, unique identifiers, valid targets, and separation between splits. Errors that violate required conditions determine whether a run can proceed; warnings are logged for investigation without necessarily stopping execution.

For FinQA, reference programs are run through the benchmark's official evaluator and checked against their expected answers. This provides an execution-based verification rather than relying solely on values recorded during annotation preparation.

FiQA validation checks that sentence text is present, entity and aspect fields are available, sentiment targets are numeric, and score values fall within the expected interval $[-1, 1]$.

E. Leakage Auditing

The exact-overlap audit compares normalized input signatures rather than depending only on record IDs. It standardizes letter case and whitespace, hashes the selected input fields, and searches for shared signatures across training, validation, and protected evaluation splits.

Potential near-duplicates are identified with word shingles and MinHash locality-sensitive hashing. Pairs with an estimated similarity threshold of 0.90 are flagged for manual inspection. They are not removed automatically, since financial documents may legitimately reuse standard wording or reporting templates.

F. Protected Evaluation Policy

Only the training partitions of FinQA and FiQA enter the training pool. Validation uses the FinQA development split and FiQA validation split. The FinQA and FiQA test partitions, along with FinBen-FOMC and FinTrust, are held back for protected evaluation.

Protected records are explicitly labelled as evaluation-only, and their identifiers are checked against the training records. Calibration choices and thresholds must be determined using validation data. Protected-set results should not be used to adjust prompts, tune hyperparameters, or select the final configuration.

TABLE II
REPORTED QLORA TRAINING CONFIGURATION.

Parameter	Value
Quantization	4-bit NF4
Double quantization	Enabled
LoRA rank	16
LoRA scaling α	32
LoRA dropout	0.05
Learning rate	2×10^{-4}
Weight decay	0.01
Warm-up ratio	0.03
Gradient clipping	1.0
Micro-batch size	1
Gradient accumulation	8
Epochs	2
Optimizer	Paged AdamW 8-bit
Random seed	42

V. METHODOLOGY

A. Task Formulation

For each example, let x_i be the financial input and y_i the corresponding reference output. The model generates $\hat{y}_i = f_\vartheta(x_i)$, with ϑ representing the pretrained parameters together with the learned adapter weights.

The FinQA prediction includes both a reasoning program and its numerical result; the program is assessed using the benchmark execution procedure. The FiQA prediction, by contrast, is a real-valued sentiment score.

Although both tasks use structured text as the training target, their outputs are evaluated according to their individual requirements. Using a common generation interface does not make their evaluation criteria interchangeable.

B. Model Configurations

The main experiment series uses Qwen3-1.7B, with Phi-4-mini-instruct included as a second model family for comparison.

Experiments cover zero-shot and few-shot prompting, fine-tuning on individual tasks, joint-task adaptation, alternative sampling schemes, and additional diagnostics. The prompting runs serve as baselines against which the effects of task-specific training can be assessed.

C. QLoRA Configuration

The reported QLoRA configuration uses four-bit NF4 quantization, double quantization, LoRA rank $r = 16$, scaling factor $\alpha = 32$, and dropout of 0.05.

The documented full-run setup permits sequences of up to 3072 tokens, although some summarized experiments may have used other limits. Each run's saved configuration should therefore be checked before differences are attributed solely to the model architecture.

D. Parameter-Efficient Adaptation

Let W_0 represent a frozen pretrained weight matrix. LoRA approximates a weight update using two lower-rank matrices:

$$W = W_0 + \Delta W, \qquad \Delta W = BA, \tag{1}$$

where A and B are trainable matrices whose inner dimension is the selected rank r.

Compared with updating the complete weight matrix, this factorization requires fewer trainable parameters. QLoRA additionally lowers memory use by quantizing the frozen base weights while learning the low-rank adapter matrices.

E. Joint-Task Training

Both datasets contribute to the joint training pool. Under balanced sampling, each dataset receives an equal sampling share; under natural sampling, the share reflects the datasets' proportions in the combined pool.

A conceptual joint objective is

$$L_{\text{joint}} = \lambda_F L_{\text{FinQA}} + \lambda_S L_{\text{FiQA}}, \tag{2}$$

where λ_F and λ_S represent the relative contribution of each task. The equation expresses the intended weighting of the two tasks. It does not, by itself, show that a dedicated numerical regression loss was implemented; that claim should only be made if the training code confirms it.

F. Experiment Workflow

Figure 1 summarizes the experimental workflow from dataset validation to protected evaluation.

VI. Experimental Protocol

A. Experiment Organization

The reported experiment series is organized as follows:

- E1–E2: zero-shot and few-shot prompting.
- E3–E4: task-specific QLoRA fine-tuning.
- E5: balanced joint QLoRA with Qwen3-1.7B.
- E6–E8: prompting and joint-training comparisons using Phi-4-mini-instruct.
- E9–E11: further prompt and sampling comparisons.
- E12: FinQA failure-category analysis.
- E13: FiQA error analysis.
- E14: confidence diagnostics.
- E15: efficiency profiling.
- E16: validation-based configuration selection.

These experiment IDs organize the analysis, but each reported metric must be traceable to its prediction artifact and saved configuration. Matching labels do not guarantee that runs used identical data, settings, or evaluation procedures.

B. Evaluation Metrics

FinQA execution accuracy measures the proportion of examples for which the predicted program executes to the reference answer under the selected evaluator. Program accuracy measures exact program agreement, while answer accuracy measures agreement of the final answer under the corresponding evaluation rule.

For FiQA, MAE is

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|. \tag{3}$$

The coefficient of determination is

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}, \tag{4}$$

where y_i is the reference score, $\hat{y}_i$ is the prediction, and $\bar{y}$ is the mean reference score.

A lower MAE means that predictions are closer to reference scores on average in absolute terms. Higher R^2 indicates a better fit than the reference-mean baseline, while a negative value means the model performs worse than that baseline on the evaluated data.

C. Validation and Model Selection

The validation split is used to choose the model configuration. Once selected, that configuration is fixed before protected evaluation begins, preventing test results from influencing model development indirectly.

The reported E5 selection score is 0.74456. The exact formula and weights used in that score should be documented in the final implementation appendix or experimental configuration. Without the formula, the score cannot be independently reconstructed from the individual task metrics.

VII. Results

A. Reported Validation Performance

Table III presents the reported validation results for the principal experiment configurations. A dash indicates that a value is not available in the current summary.

The reported runs show a clear difference between the prompting baselines and the fine-tuned systems. E1 and E2 have very low FinQA execution accuracy, while the fine-tuned configurations report higher values. Within this experimental setup, adaptation appears useful for producing the structured programs required by FinQA.

The zero-valued entries should remain unchanged until the underlying metric exports have been inspected. The evaluator should separately record malformed outputs, parsing failures, and valid predictions that are simply incorrect.

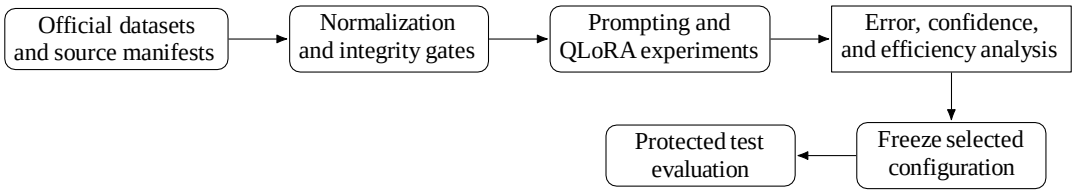


Fig. 1. Research workflow. Model selection uses training and validation data; protected evaluation follows configuration freezing.

TABLE III
REPORTED VALIDATION RESULTS. THESE VALUES ARE NOT PROTECTED-TEST RESULTS.

ID	Configuration	Exec. Acc.	Prog. Acc.	Ans. Acc.	FiQA MAE	FiQA R^2	Latency (s)
E1	Qwen3 zero-shot	0.000	0.000	0.005	0.501	-1.787	3.40
E2	Qwen3 few-shot	0.000	0.000	0.000	0.185	0.472	3.08
E3	Qwen3 FinQA-only QLoRA	0.681	0.624	0.390	0.505	-1.840	1.19
E4	Qwen3 FiQA-only QLoRA	0.000	0.000	0.000	0.129	0.725	2.35
E5	Qwen3 joint balanced QLoRA	0.675	0.612	0.376	0.1301	0.7301	1.18
E6	Phi zero-shot	0.000	0.000	0.000	0.381	-0.444	3.34
E7	Phi few-shot	0.040	0.034	0.006	0.325	-0.181	3.53
E8	Phi joint balanced QLoRA	0.701	0.640	0.386	0.130	0.809	1.08
E11	Qwen3 joint natural sampling	0.6149	0.555	0.344	0.1276	0.7355	1.20

B. Joint Training and Sampling

E5 reports execution accuracy of 0.6750 and FiQA MAE of 0.1301. E11 reports execution accuracy of 0.6149 and FiQA MAE of 0.1276.

Thus, the reported balanced configuration has higher FinQA execution accuracy, whereas the natural-sampling configuration has slightly lower FiQA MAE. The result is consistent with a possible trade-off between the two task objectives, but it does not establish statistical significance.

Claims about the general superiority of balanced sampling would require repeated runs with multiple random seeds and measures of variation. A fair comparison should also control, as far as possible, for dataset proportions, task sampling, truncation settings, and the number of optimization steps.

C. Secondary Model Comparison

E8, using Phi-4-mini-instruct, reports execution accuracy of 0.701 and FiQA R^2 of 0.809. These values are higher than the corresponding E5 values in the reported summary.

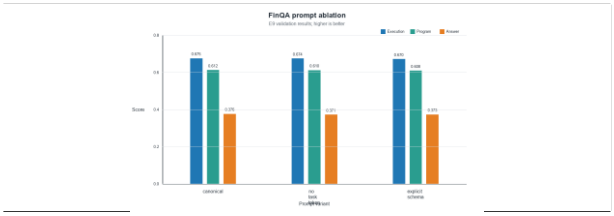
A conclusion about model architecture would require the two systems to use comparable datasets, prompts, training budgets, sequence limits, and evaluation code. Until these factors are verified, E8 is best viewed as an encouraging result, not definitive evidence that one model family is superior.

D. Prompt Ablation

The prompt variants produce similar reported execution-accuracy values. Removing the task token causes only a small change, while adding an explicit schema produces a somewhat lower score than the canonical prompt. This may suggest sensitivity to prompt design, although repeated experiments or uncertainty estimates are needed to determine whether the differences matter in practice.

TABLE IV
REPORTED FINQA PROMPT ABLATIONS.

Prompt variant	Execution accuracy
Canonical	0.6750
No task token	0.6738
Explicit schema	0.6704



For FiQA, the explicit-schema variant reports MAE of 0.1273 and $R^2 = 0.7378$, compared with canonical MAE of 0.1301 and $R^2 = 0.7301$. The reported values suggest a modest improvement in regression performance for this variant. They should be regenerated from the canonical predictions before publication.

VIII. FAILURE ANALYSIS

An incorrect numerical response can arise for several reasons: the model may retrieve the wrong evidence, generate an invalid program, choose an unsuitable operation, calculate incorrectly, or report a final value inconsistent with program execution. Overall accuracy therefore provides only a partial view and should be accompanied by failure categories.

The reported E12 summary is shown in Table V.

TABLE V
REPORTED E12 FINQA FAILURE CATEGORIES.

Category	Count	Percentage
Correct outcomes	304	34.4%
Reasoning/execution errors	283	32.0%
Program correct, answer wrong	265	30.0%
Equivalent alternate programs	27	3.1%
Invalid programs	3	0.3%
Answer correct, program wrong	1	0.1%
Total	883	100%

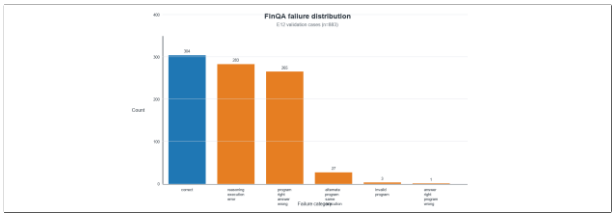


Fig. 3. Audited FinQA failure-category distribution for E5 validation predictions.

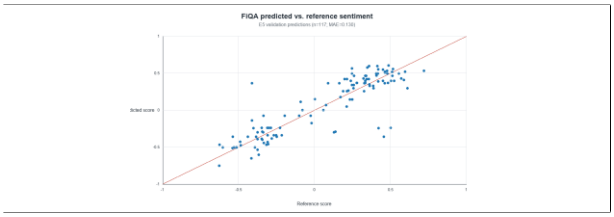


Fig. 4. Reference versus predicted FiQA sentiment scores for E5 validation predictions.



Fig. 5. Absolute-error distribution for FiQA E5 validation predictions.

A. *Metric Consistency Audit*

A material inconsistency is present in the available results. E12 reports 304 correct outcomes out of 883 examples, approximately 34.4%. E5 reports FinQA execution accuracy of 0.6750.

The two figures should not be assumed to measure the same outcome on the same examples. To determine whether E12 belongs to a different run or reflects a reporting problem, the team needs to inspect the saved predictions, run IDs, evaluator versions, and metric definitions.

Until the discrepancy is resolved, these values should not support a unified conclusion. Both the overall metrics and the failure counts should be recomputed from one canonical evaluation export.

B. *Implications for Financial Reasoning*

Failure labels can help identify where improvements are needed. Invalid programs may call for stricter output formatting or validation; arithmetic errors warrant separating calculation checks from text generation; and cases with a correct answer but a different program show why program exact-match and final-answer correctness should be reported separately.

For practical deployment, recognizing invalid outputs and abstaining when necessary may be as important as improving aggregate accuracy. Any rejection rule should be assessed on data independent of the examples used to design or tune it.

IX. FIQA ERROR ANALYSIS

Sentiment regression should be assessed through both summary metrics and the shape of individual predictions. Even when MAE appears acceptable, a model may consistently pull strongly positive or negative examples toward the middle of the score range.

The reported E5 result has MAE of 0.1301 and R^2 = 0.7301. E11 reports MAE of 0.1276 and R^2 = 0.7355. E11 has a marginally smaller absolute error and a marginally higher coefficient of determination in the reported results. The size of the validation split and run-to-run variation should be considered before treating these differences as meaningful.

Useful diagnostics include reference-versus-predicted scatter plots, residual distributions, error summaries by sentiment range, and inspection of examples with the largest absolute errors.

A. *Error Interpretation*

When predicted scores closely follow the diagonal in a reference-versus-prediction plot, the model's outputs generally agree with the annotations. Persistent deviations can reveal systematic bias, compression toward the mean, or errors introduced while parsing generated numbers.

Residual checks should separate mistakes in the model's prediction from formatting and parsing failures. Out-of-range values should be counted and disclosed; they should not be silently clipped unless clipping was defined in advance as part of the evaluation procedure.

X. CONFIDENCE DIAGNOSTICS

A confidence score is informative only if it relates meaningfully to the chance of correctness or to the size of a prediction error. The likelihood assigned to generated tokens cannot automatically be interpreted as a calibrated confidence estimate.

The reported E14 diagnostics give a Spearman correlation of 0.125 between FinQA confidence and correctness, with ROC-AUC of 0.577. For FiQA, the reported Spearman correlation between confidence and absolute error is 0.237.

The reported correlations suggest only a weak ranking relationship in this diagnostic setup. They do not show that the model produces calibrated probabilities or that a chosen threshold can reliably flag incorrect responses.

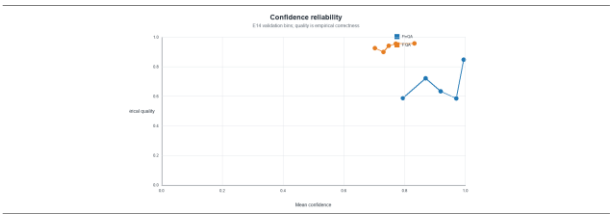


Fig. 6. Confidence reliability by validation confidence bin for FinQA and FiQA.



Fig. 7. Reported final training loss after two epochs for E5 and E8; per-step logs were not exported.

TABLE VI
REPORTED E5 EFFICIENCY MEASUREMENTS.

Measurement	Reported value
Training examples	7073
Epochs	2
Training duration	2.47 hours
Peak VRAM	3.40 GiB
Final training loss	0.350
Inference latency	1.18 s/item
GPU	RTX 4050 Laptop, 6 GB

A. Calibration and Selective Prediction

A follow-up calibration experiment should fit any confidence mapping and choose abstention thresholds using validation data only. Depending on how confidence and the target event are defined, useful diagnostics may include reliability plots, expected calibration error, Brier score, and negative log-likelihood.

In a selective-prediction setting, coverage is the proportion of inputs for which the system provides an answer, while selective accuracy measures correctness among those accepted outputs. A risk-coverage curve can summarize the balance between answering a larger share of questions and maintaining prediction quality.

Such metrics are meaningful only when confidence and correctness have precise definitions. For regression, confidence should be linked to expected prediction error in a defensible way instead of being derived uncritically from token probabilities.

XI. EFFICIENCY ANALYSIS

Computational cost is relevant because the target use case includes environments with limited resources. For E5, the reported run processed 7,073 training records for two epochs, took 2.47 hours, and reached a peak GPU-memory use of 3.40 GiB on an NVIDIA RTX 4050 Laptop GPU with 6 GB of memory. The run's reported final training loss was 0.350, and average inference latency was 1.18 seconds per item.

These measurements need to be accompanied by the conditions under which they were obtained. Latency can vary with prompt length, output length, batch size, device load, and decoding parameters. Memory use is likewise affected by sequence length, quantization, optimizer state, and implementation choices.

A. Reproducible Performance Profiling

For efficiency results to be reproducible, the report should state the warm-up count, number of timed examples, batch size, input-length distribution, generation limit, and whether tokenization and data transfer are included in the timing.

Training duration and inference latency measure different things and should not be compared directly. Training duration reflects the cost of adapting the model, whereas inference latency measures prediction time under a specified serving setup.

XII. DISCUSSION

Taken at face value, the reported experiments provide a useful baseline for applying parameter-efficient fine-tuning to financial reasoning and continuous sentiment prediction. The gap between prompting and fine-tuned runs suggests that adapting to the structured targets is beneficial in this particular setup.

The two sampling approaches show different task-level results: E5 has higher reported FinQA execution accuracy, while E11 has a slightly lower FiQA MAE. Repeated experiments under matched training budgets are needed before making a broad claim about which sampling strategy is preferable.

The Phi-4-mini-instruct run also produces encouraging validation figures. However, architecture-level comparisons require a controlled setup because sequence limits, prompt construction, optimization settings, and decoding choices can all affect the results.

The confidence results reinforce that average accuracy is not enough to establish deployment reliability. A model may perform reasonably on average while remaining unable to identify its own mistakes. Calibration and selective prediction therefore need independent evaluation; they should not be presumed to follow from fine-tuning.

A. Research Integrity

Phase-2.1 improves traceability by recording source fingerprints, checking FinQA reference programs, and auditing exact and approximate overlap between splits. These safeguards make the experiment easier to reproduce and help reduce the possibility that accidental contamination inflates a score.

These controls do not remove all validity concerns. Potential near-duplicates still need human review, metrics must be computed consistently between runs, and file hashes cannot establish whether an annotation is factually correct.

B. Limitations

Several limitations constrain the conclusions that can be drawn from this study.

First, FiQA validation includes only 117 examples, making its regression metrics potentially sensitive to a small number of records. Second, the available summary contains no multi-seed significance analysis. Third, differences in sequence length and other configuration details may confound comparisons between runs.

Fourth, the disagreement between the E5 and E12 FinQA results has not yet been resolved. Fifth, the confidence analysis does not demonstrate calibrated uncertainty. Sixth, this report does not include protected-test results. Finally, the use and coverage of official evaluators for all seven FinTrust dimensions must be verified before claiming a complete trustworthiness assessment.

XIII. Protected Evaluation and Future Work

The immediate next step is to investigate the E5/E12 mismatch. The original predictions should be processed again with the official FinQA executor, after which execution accuracy, program accuracy, answer accuracy, and failure-category totals should all be calculated from the same canonical export.

Once the metrics have been verified, the chosen model and its settings should be locked. Final evaluation can then be conducted on the FinQA and FiQA test partitions, FinBen-FOMC, and the available FinTrust dimensions using the relevant evaluation procedures.

FinQA reporting should include the official execution result together with clearly specified program- and answer-level metrics. FiQA should be assessed using MAE, MSE, R^2, and suitable correlation measures. For FinBen-FOMC, report accuracy, macro-F1, and class-level F1 scores. FinTrust should retain separate, official procedures for its individual dimensions rather than reducing them to one undifferentiated number.

Calibration methods and abstention thresholds should be selected solely on validation data. The protected test sets should then be reserved for final reporting and should not be used to make further prompt or hyperparameter changes.

Further experiments should include several random seeds, uncertainty intervals, finer-grained error labels, and controlled comparisons across model families. Such evidence would help determine whether the observed patterns extend beyond the current validation samples.

XIV. Conclusion

This paper describes a traceable baseline for adapting small language models to financial numerical reasoning and continuous sentiment regression. The workflow brings together QLoRA fine-tuning, data-integrity validation, protected-set controls, prompt and sampling experiments, error analysis, confidence diagnostics, and measurements of computational cost.

The available validation figures suggest that parameter-efficient fine-tuning can outperform the reported prompting baselines on the structured financial tasks examined here. Results from balanced and natural sampling hint at a trade-off between numerical reasoning and sentiment regression, and the secondary model-family run motivates a more controlled comparison.

These observations remain provisional because the E5/E12 inconsistency has not been resolved. Final claims should rely on metrics regenerated from canonical predictions and on a subsequent protected-test evaluation. The proposed workflow supports that verification process, but it does not independently demonstrate calibrated uncertainty or establish that autonomous financial decisions would be safe.

Acknowledgment

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